

Assignment 1

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Q1: [20] Three identical charges of charge Q are located at the vertices of an equilateral triangle of side length a . Determine the force on one of the charges due to the other two.

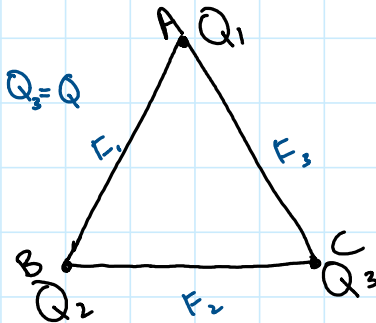
By Coulomb's law:-

$$F_1 = \frac{Q^2}{4\pi\epsilon_0 a^2}$$

as triangle is equilateral so,

$$F_1 = F_2 = F_0$$

$$Q_1 = Q_2 = Q_3 = Q$$



• Vector addition:-

$$F_{\text{net}} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos(60^\circ)}$$

Now,

$$F_0 = \frac{Q^2}{4\pi\epsilon_0 a^2} \quad \because \cos(60^\circ) = 1/2$$

$$\begin{aligned} F_{\text{net}} &= \sqrt{F_0^2 + F_0^2 + 2F_0^2 \left(\frac{1}{2}\right)} \\ &= \sqrt{3F_0^2} = \sqrt{3} F_0 \end{aligned}$$

$$F_{\text{net}} = \frac{\sqrt{3} Q^2}{4\pi\epsilon_0 a^2} \hat{n} \quad \text{Answer.}$$

with vector,
angle bisector at
vertex 'C'

Q2: [40] A total charge of Q_1 is distributed uniformly along a semicircular ring as shown in Figure 1. Two point charges, each of magnitude Q_2 , are situated as shown. The surrounding medium is free space.

- (a) Find Q_2 in terms of Q_1 so that the potential Φ at the center of the ring is zero.
- (b) Find Q_2 in terms of Q_1 so that the electric field $\mathbf{E}$ at the center of the ring is zero.

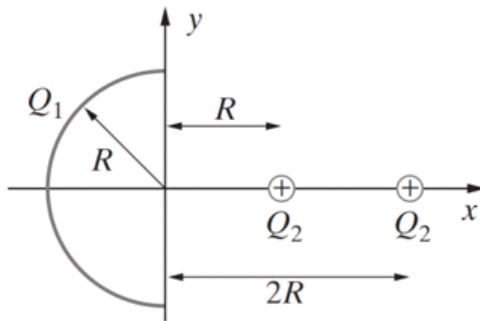


Figure 1: Figure for Question 2.

• Part (a):

$V=0$ at center $(0,0)$

1- potential due to semicircular Ring (Q_1):

$$V_{\text{ring}} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{R} = \frac{Q_1}{4\pi\epsilon_0 R}$$

2- potential due to point charge (Q_2):

$$V_{\text{point}} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_2}{R} + \frac{Q_2}{2R} \right) = \frac{3Q_2}{8\pi\epsilon_0 R}$$

3- Total potential:

$$V_{\text{total}} = V_{\text{ring}} + V_{\text{point}} = \frac{1}{4\pi\epsilon_0 R} \left(Q_1 + \frac{3}{2} Q_2 \right) = 0$$

$$Q_1 + \frac{3}{2} Q_2 = 0 \Rightarrow Q_2 = -\frac{2}{3} Q_1$$

$$Q_1 + \frac{3}{2} Q_2 = 0 \Rightarrow Q_2 = -\frac{2}{3} Q_1$$

• Part (b): electric field $E=0$ at center $(0,0)$

1- electric field due to semicircular Ring (Q_1):

$$x \leq 0 \quad \phi = \pi/2 \text{ to } \phi = 3\pi/2$$

$$\therefore \lambda = \frac{Q_1}{\pi R}$$

$$E_{x, \text{ring}} = \int_{\pi/2}^{3\pi/2} \frac{\lambda (R d\phi)}{4\pi \epsilon_0 R^2} (-\cos \phi) = \frac{-\lambda}{4\pi \epsilon_0 R} \int_{\pi/2}^{3\pi/2} \cos \phi d\phi$$

$$\text{since: } \int_{\pi/2}^{3\pi/2} \cos \phi d\phi = -1 - 1 = -2$$

$$E_{x, \text{ring}} = \frac{2\lambda}{4\pi \epsilon_0 R} = \frac{2(Q_1/\pi R)}{4\pi \epsilon_0 R} = \frac{Q_1}{2\pi^2 \epsilon_0 R^2}$$

$$\vec{E}_{\text{ring}} = \frac{Q_1}{2\pi^2 \epsilon_0 R^2} \hat{a}_x$$

2- electric field due to point charges (Q_2)

x -axis at $(R,0)$ and $(2R,0)$

$$= \frac{Q_2}{4\pi \epsilon_0 R^2} (-\hat{a}_x) + \frac{Q_2}{4\pi \epsilon_0 (2R)^2} (-\hat{a}_x)$$

$$= -\frac{Q_2}{4\pi \epsilon_0 R^2} \left(1 + 1/4\right) \hat{a}_x = -\frac{5Q_2}{16\pi \epsilon_0 R^2} \hat{a}_x$$

3- Total electric field setting to zero.

$$E_{\text{total}} = \left(\frac{Q_1}{4\pi^2 \epsilon_0 R^2} - \frac{5Q_2}{16\pi \epsilon_0 R^2} \right) \hat{a}_x = 0$$

$$\frac{Q_1}{4\pi^2 \epsilon_0 R^2} = \frac{5Q_2}{16\pi \epsilon_0 R^2} \Rightarrow Q_2 = \frac{16\pi}{10\pi^2} Q_1 = \boxed{\frac{8}{5\pi} Q_1} \text{ Ans.}$$

Q3: [40] The space between two concentric spheres of radii a and b ($a < b$) in free space is charged to a volume charge density given by

$$\rho(r) = \frac{K}{r^2}; \quad a \leq r \leq b$$

where K is a constant. (a) Find the total charge in the shell. (b) Find the electric field at all points in space. (c) Find the electric potential at all points in space. (d) What happens if $b \rightarrow a$?

a) Total charge:

$$Q_{\text{total}} = \int_a^b \rho(r) 4\pi r^2 dr = \int_a^b \frac{K}{r^2} (4\pi r^2) dr = 4\pi K \int_a^b dr = 4\pi K(b-a)$$

b) Electric field $E(r)$:

1- $r < a$:

$$(Q_{\text{enc}} = 0) \\ E(r) = 0$$

2- region $a \leq r \leq b$:

$$Q_{enc}(r) = \int_a^r \frac{k}{r'^2} (4\pi r'^2) dr' = 4\pi k(r-a)$$

$$E_r(r) \cdot 4\pi r^2 = \frac{4\pi k(r-a)}{\epsilon_0}$$

$$E(r) = \frac{k(r-a)}{\epsilon_0 r^2} \hat{a}_r$$

3- region $r > b$:

$$Q_{enc} = Q_{total} = 4\pi k(b-a)$$

$$E_r(r) \cdot 4\pi r^2 = \frac{4\pi k(b-a)}{\epsilon_0}$$

$$E(r) = \frac{k(b-a)}{\epsilon_0 r^2} \hat{a}_r$$

c) Electric Potential $V(r)$ at all points in space.

1- region $r \geq b$

$$V(r) = - \int_{\infty}^r \frac{k(b-a)}{\epsilon_0 r'^2} dr' = \frac{k(b-a)}{\epsilon_0 r}$$

2- Region $a \leq r \leq b$

$$V(r) = V(b) - \int_b^r E_r(r') dr'$$

$$= \frac{k(b-a)}{\epsilon_0 b} - \int_b^r \frac{k(r'-a)}{\epsilon_0} dr'$$

$$V(r) = \frac{k(b-a)}{\epsilon_0 b} - \frac{k}{\epsilon} \left[\ln\left(\frac{r}{b}\right) + a\left(\frac{1}{r} - \frac{1}{b}\right) \right]$$

$$= \frac{k}{\epsilon} \left[1 - a/r - \ln(r/b) \right]$$

3- region $r < a$:

since $E=0$, $V(r)$ is constant:

$$\begin{aligned} V(r) = V(a) &= \frac{k}{\epsilon_0} \left[1 - 1 - \ln(a/b) \right] \\ &= \frac{k}{\epsilon} \ln(b/a) \end{aligned}$$

d) limit $b \rightarrow a$:

- $Q_{\text{total}} = 4\pi k(b-a) \rightarrow 0$
- $E(r) \rightarrow 0$ everywhere in space
- $V(r) \rightarrow 0$ everywhere in space.

[25 points] Q4: Applying Coulomb's law, find the electric field at point P with Cartesian coordinates $(0, 0, h)$ due to a circular disk of radius a residing in the x - y plane, and assume that the surface charge density to vary with r as

$$\rho_s = c\rho^3 \quad (c > 0) \quad (\text{C/m}^2)$$

- Some position vector: $\mathbf{r}' = \rho' \hat{\mathbf{a}}_\rho$
- Let $\mathbf{r}$ be the position vector from the origin to point P .

- Source position vector: $\mathbf{r}' = \rho' \hat{\mathbf{a}}_\rho$
- field position vector: $\mathbf{r} = h \hat{\mathbf{a}}_z$
- Displacement vector: $\mathbf{R} = \mathbf{r} - \mathbf{r}' = -\rho' \hat{\mathbf{a}}_\rho + h \hat{\mathbf{a}}_z$

- Distance $R = |\mathbf{R}| = \sqrt{(\rho')^2 + h^2}$

- Differential surface elements: $ds' = \rho' d\rho' d\phi'$

2- Integral formula:

$$E = \frac{1}{4\pi\epsilon_0} \int \frac{\rho_s(\rho') ds'}{R^3} \mathbf{R}$$

$$= \frac{C}{4\pi\epsilon} \int_0^{2\pi} d\phi' \int_0^a \frac{(\rho')^3 (\rho' d\rho')}{[(\rho')^2 + h^2]^{3/2}} (-\rho' \hat{\mathbf{a}}_\rho + h \hat{\mathbf{a}}_z)$$

By azimuthal symmetry:
 $-\rho' \hat{\mathbf{a}}_\rho$ over $\phi' \in [0, 2\pi]$ vanishes

Integrating $d\phi'$ gives 2π :

$$E = \frac{Ch}{2\epsilon_0} \left(\int_0^a \frac{(\rho')^4}{[(\rho')^2 + h^2]^{3/2}} d\rho' \right) \hat{\mathbf{a}}_z$$

3- Evaluating the integral:

$$I = \int_0^a \frac{x^4}{(x^2 + h^2)^{3/2}} \cdot dx$$

$$u = x^2 \quad dv = \frac{x}{(x^2 + h^2)^{3/2}} dx$$

$$u = x^3 \quad dv = \frac{x}{(x^2+h^2)^{3/2}} dx$$

$$I = \left[\frac{-x^3}{\sqrt{x^2+h^2}} \right]_0^a + 3 \int_0^a \frac{x^2}{\sqrt{x^2+h^2}} dx$$

$$I = \frac{-a^3}{\sqrt{a^2+h^2}} + \frac{3a}{2} \sqrt{a^2+h^2} - \frac{3h^2}{2} \ln \left(\frac{a + \sqrt{a^2+h^2}}{h} \right)$$

Combining all terms:-

$$\frac{-a^3}{\sqrt{a^2+h^2}} + \frac{3a(a^2+h^2)}{2\sqrt{a^2+h^2}} = \frac{a(a^2+3h^2)}{2\sqrt{a^2+h^2}}$$

Now,

$$\boxed{E = \frac{ch}{2\epsilon_0} \left[\frac{a(a^2+3h^2)}{2\sqrt{a^2+h^2}} - \frac{3h^2}{2} \ln \left(\frac{a + \sqrt{a^2+h^2}}{h} \right) \right] \hat{a}_z}$$

[25 points] Q5: In a certain region of space, the electric field is given in spherical coordinates as follows:

$$\mathbf{E} = \frac{c}{r^4} \mathbf{a}_r \quad (\text{V/m}) \quad (c > 0)$$

Applying Coulomb's law, find the electric potential of point A with respect to point B where A is at +2 m and B at -4 m, both on the x-axis.

Hint: Since both field points are on x-axis, therefore, one may write

$$\mathbf{E} = \pm \frac{20}{x^4} \mathbf{a}_x \quad (\text{V/m})$$

$$E = -\nabla V \rightarrow E_r = -\frac{dV}{dr} = \frac{c}{r^4}$$

$$V(r) = - \int \frac{c}{r^4} dr = \frac{c}{3r^3}$$

$$V(r) = - \int \frac{C}{r^3} dr = \frac{C}{3r^2}$$

2- radial distance:

$$\text{Point (A)} [2, 0, 0], r_A = |+2| = 2 \text{ m}$$

$$\text{Point (B)} [-4, 0, 0], r_B = |-4| = 4 \text{ m}$$

• Potential difference:-

$$V_{AB} = V(A) - V(B)$$

$$V(A) = \frac{C}{(2)^3 \cdot 3} = \frac{C}{24} \text{ V}$$

$$V(B) = \frac{C}{3(4)^3} = \frac{C}{192} \text{ V}$$

$$V_{AB} = V(A) - V(B) = \frac{C}{24} - \frac{C}{192} = \boxed{\frac{7C}{192} \text{ V}}$$

[25 points] Q6: In a certain region of space, the volumetric charge density is given in cylindrical coordinates by the function:

$$\rho_v = c \rho e^{-\rho} \quad (c > 0) \quad (\text{C/m}^3)$$

(a) Apply the integral form of Gauss's law $\oint_S \mathbf{D} \cdot d\mathbf{s} = Q_{\text{enclosed}}$ to find $\mathbf{D}$.

(b) Apply the differential form of Gauss's law $\nabla \cdot \mathbf{D} = \rho_v$ to find $\mathbf{D}$.

Hint: In both cases, one may assume $\mathbf{D} = D_\rho \mathbf{a}_\rho$.

a) from gauss's law:- $\mathbf{D} = D_\rho(\rho) \hat{\mathbf{a}}_\rho$

∴ • from gauss's law:- $D = D_p(\rho) \hat{a}_\rho$

$$\text{electric flux} = \oint_S D \cdot d\mathbf{s} = D_p(\rho) \cdot (2\pi\rho L)$$

• enclosed charge:

$$\begin{aligned} Q_{enc} &= \int_0^L dz \int_0^{2\pi} d\phi \int_0^\rho (c\rho' e^{-\rho'}) \rho' d\rho' \\ &= 2\pi cL \int_0^\rho \rho'^2 e^{-\rho'} d\rho' \end{aligned}$$

∴ using integration by parts.

$$\int x^2 e^{-x} \cdot dx = -e^{-x} (x^2 + 2x + 2)$$

$$Q_{enc} = 2\pi cL [2 - e^{-\rho} (\rho^2 + 2\rho + 2)]$$

• equating Flux and enclosed charge:-

$$D_p(2\pi\rho L) = 2\pi cL [2 - e^{-\rho} (\rho^2 + 2\rho + 2)]$$

$$D = \frac{c}{\rho} [2 - e^{-\rho} (\rho^2 + 2\rho + 2)] \hat{a}_\rho$$

b)

differential form of gauss's law:-

1- Divergence in cylindrical coordinate:

$$\nabla \cdot \mathbf{D} = \frac{1}{\rho} \frac{d}{d\rho} (\rho D_\rho) = \rho_v = c \rho e^{-\rho}$$

2- Integration-

$$\frac{d}{d\rho} (\rho D_\rho) = c \rho^2 e^{-\rho}$$

$$\rho D_\rho = c \int \rho^2 e^{-\rho} d\rho = -c e^{-\rho} (\rho^2 + 2\rho + 2) + C_1$$

3- Boundary Condition:-

$$\rho = 0, \quad \rho D_\rho = 0:$$

$$0 = -C(2) + C_1 \rightarrow C_1 = 2c$$

$$\rho D_\rho = c [2 - e^{-\rho} (\rho^2 + 2\rho + 2)]$$

$$\mathbf{D} = \frac{c}{\rho} [2 - e^{-\rho} (\rho^2 + 2\rho + 2)] \hat{\mathbf{a}}_\rho$$

[25 points] Q7: A semi-circular ring of radius a carries a uniform line charge density ρ_L [C/m] and is placed on the xy -plane with axis the same as the z -axis. Obtain the electric field intensity $\mathbf{E}$ at the origin $(0, 0, 0)$ by using the following expression:

$$\mathbf{E}(\mathbf{r}) = \int_L \frac{\rho_L d\mathbf{l}' (\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3}$$

and show that

$$\mathbf{E}(0, 0, 0) = -\frac{\rho_L}{2\pi\epsilon_0 a} \mathbf{a}_y$$

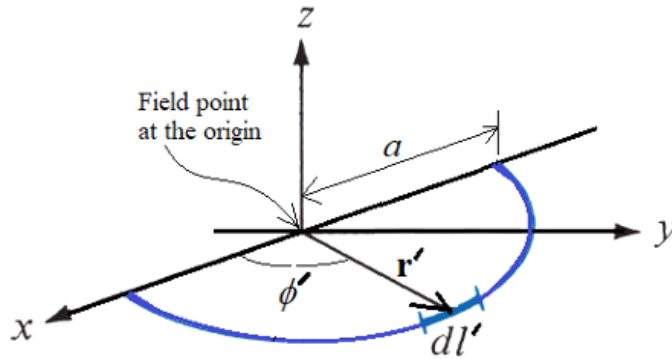


Figure 1: $\mathbf{E}$ field due to a semi-circular ring of charge.

Start with specifying the position vectors $\mathbf{r}$ and $\mathbf{r}'$ for the field point and (differential) source element, respectively, and continue to discuss the detailed evaluation of integral function involved in this problem. Once you have a complete expression of integral function, you may happen to drop one or more terms owing to any particular symmetry in the charge configuration.

• Position vectors:-

- $\mathbf{r} = (0, 0, 0) = \mathbf{0}$

- $\mathbf{r}' = a \cos \phi' \hat{\mathbf{a}}_x + a \sin \phi' \hat{\mathbf{a}}_y$

- displacement vector $\mathbf{r} - \mathbf{r}' = -a \cos \phi' \hat{\mathbf{a}}_x - a \sin \phi' \hat{\mathbf{a}}_y$

- Distance $|\mathbf{r} - \mathbf{r}'| = a$

- Differential length element: $d\mathbf{l}' = a d\phi'$

• Integral expression

• Integral expression

$$E(0,0,0) = \int_0^\pi \frac{\rho_L (a d\phi) (-a \cos \phi' \hat{a}_x - a \sin \phi' \hat{a}_y)}{4\pi \epsilon_0 a^3}$$

$$E(0,0,0) = \frac{\rho_L}{4\pi \epsilon_0 a} \left[\left(\int_0^\pi -\cos \phi' d\phi' \right) \hat{a}_x + \left(\int_0^\pi -\sin \phi' d\phi' \right) \hat{a}_y \right]$$

Now,

• x-component-

$$\int_0^\pi -\cos(\phi') d\phi' = -\sin \phi' \Big|_0^\pi = 0$$

• y-component-

$$\int_0^\pi -\sin \phi' d\phi' = \cos \phi' \Big|_0^\pi = -1 - 1 = -2$$

Now,

$$E(0,0,0) = \frac{\rho_L}{4\pi \epsilon_0 a} (-2\hat{a}_y) = \boxed{\frac{-\rho_L}{2\pi \epsilon_0 a} \hat{a}_y} \text{ Answer.}$$